

COMP 2121 – Quiz 1

Time: 20 min.

Possible marks: 5

Name:

Set:

Student #:

Note: You do not need to simplify your answers unless specified otherwise.

1. Consider all octal strings of length 12.

digits 0,1,2,3,4,5,6,7

a) How many strings contain at least one 0? (1 mark)

all - no 0s

$$8^{12} - 7^{12}$$

b) How many strings have exactly five 1's, exactly six 2's, and one other digit? (1 mark)

$$\frac{12!}{5!6!} \cdot 6$$

select other digit
shuffle 22222 11111 -

c) How many strings with exactly five 1's, exactly six 2's and one other digit have no adjacent 1's? (1 mark)

2 2 2 2 2 2 -

insert 5 1's

$$\binom{8}{5} \cdot \frac{7!}{6!} \cdot 6$$

select other digit
arrange 22222 -

Formulae: $P(n, r) = \frac{n!}{(n-r)!}$

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$

2. What value does the variable *counter* have after the following code execution? (2 marks)
(Note: the *for* loops include both the start and end values)

```

counter = 500

for i = 3 to 52 {
    counter = counter + 2
}
for j = 1 to n {
    counter = counter + 3
    for k = j+1 to 2n+2 {
        counter = counter + 5
    }
}
// assume n>0

```

$$i: 52 - 3 + 1 = 50$$

gap: no pairs made of $n+1, n+2, \dots, 2n+2$

$$\text{Counter} = 500 + \underline{50} (2) + \underline{n} (3) + \underline{\left[\binom{2n+2}{2} - \binom{n+2}{2} \right]} (5)$$

Formulae: $P(n, r) = \frac{n!}{(n-r)!}$

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r! (n-r)!}$$

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r! (n-1)!}$$